蔡老师班补充作业6

试卷4：卅三（2021年）

$x-2y+3z+2=0 \Rightarrow \vec{n_1}=(1,-2,3)$

$x+2y-3z-2=0 \Rightarrow \vec{n_2}=(1,2,-3)$

$\Rightarrow \vec{n}=\vec{n_1}\times\vec{n_2}=\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -2 & 3 \\ 1 & 2 & -3 \end{vmatrix}=6\vec{j}+4\vec{k}$

$P(1,0,1) \Rightarrow$ the plane: $6(y-0)+4(z-1)=0 \Rightarrow \underline{3y+2z=2}$

试卷三：一(1)(3)；三(3)

(1) ✓ $r=2\sin\theta \Rightarrow r^2=2r\sin\theta \Rightarrow x^2+y^2=2y \Rightarrow x^2+(y-1)^2=1$

(3). ✓ $\vec{u}\times\vec{v}=\vec{u}\times\vec{w} \Rightarrow \vec{u}\times(\vec{v}-\vec{w})=\vec{0} \Rightarrow \vec{u}\parallel(\vec{v}-\vec{w})$

$\vec{u}\cdot\vec{v}=\vec{u}\cdot\vec{w} \Rightarrow \vec{u}\cdot(\vec{v}-\vec{w})=0 \Rightarrow \vec{u}\perp(\vec{v}-\vec{w})$

$\Rightarrow \vec{v}-\vec{w}=\vec{0}$

$\Rightarrow \vec{v}=\vec{w}$.

三(3) $\begin{cases} x=-1+2t \\ y=3+t \\ z=-t \end{cases} \Rightarrow \vec{v_1}=(2,1,-1)$ $\quad \begin{cases} x=-2t \\ y=t \\ z=1-t \end{cases} \Rightarrow \vec{v_2}=(-2,1,-1)$

$\Rightarrow \vec{n}=\vec{v_1}\times\vec{v_2}=\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 1 & -1 \\ -2 & 1 & -1 \end{vmatrix}=4\vec{j}+4\vec{k}$

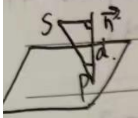
取 $P(-1,3,0) \Rightarrow 4(y-3)+4z=0 \Rightarrow \underline{y+z=3}$

chapter 12 # 8, 17, 18.

8. $S(x_0, y_0, z_0)$, plane: $Ax+By+Cz+D=0$.

$\vec{n}=(A, B, C)$

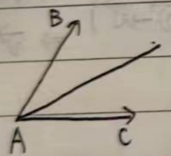
取平面上一点 $P(x, y, z)$, $Ax+By+Cz+D=0$



$$\Rightarrow d=\frac{|\vec{PS}\cdot\vec{n}|}{|\vec{n}|}=\frac{|A(x-x_0)+B(y-y_0)+C(z-z_0)|}{\sqrt{A^2+B^2+C^2}}$$

$$=\frac{|Ax_0+By_0+Cz_0+D|}{\sqrt{A^2+B^2+C^2}}$$

17. $\vec{AB}=(-3, 0, 4)$, $\vec{AC}=(5, -2, -14)$



$$\Rightarrow \vec{AD}=\frac{\vec{AB}}{|\vec{AB}|}+\frac{\vec{AC}}{|\vec{AC}|}=\frac{1}{5}(-3, 0, 4)+\frac{1}{15}(5, 2, -14)$$

$$=\left(\frac{-4}{15}, -\frac{2}{15}, -\frac{2}{15}\right)$$

$$\Rightarrow |\vec{AD}|=\sqrt{\frac{1}{15^2}(16+4+4)}=\sqrt{\frac{8}{15}}=\frac{2}{5}\sqrt{\frac{2}{3}}$$

$$\Rightarrow \frac{\vec{AD}}{|\vec{AD}|}=\frac{1}{\frac{2}{5}\sqrt{\frac{2}{3}}}\left(-\frac{4}{15}, \frac{2}{-15}, \frac{2}{-15}\right)$$

$$=\left(-\sqrt{\frac{2}{3}}, -\frac{1}{2}\sqrt{\frac{2}{3}}, -\frac{1}{2}\sqrt{\frac{2}{3}}\right)$$

$$=\left(-\frac{2}{\sqrt{6}}, \frac{-1}{\sqrt{6}}, \frac{-1}{\sqrt{6}}\right)$$

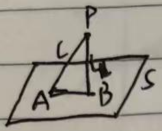
18. (a) l: $P(1, -1, 0)$, $\vec{v_1}=(1, -1, 1)$; S: $\vec{n_1}=(3, -1, 1)$

$$\Rightarrow \vec{n}=\vec{v_1}\times\vec{n_1}=\begin{vmatrix}\vec{i} & \vec{j} & \vec{k}\\ 1 & -1 & 1\\ 3 & -1 & 1\end{vmatrix}=2\vec{j}+2\vec{k}$$

$\Rightarrow$ plane: $2(y+1)+2z=0$

$\Rightarrow y+z+1=0$

(b). l与平面S的交点A: $3(1+t)-(-1-t)+t=5 \Rightarrow 5t=1 \Rightarrow t=\frac{1}{5}$



$A\left(\frac{6}{5}, -\frac{6}{5}, \frac{1}{5}\right)$

取l上点$P(1,-1,0)$, 过点P且垂直S的直线l_1: $\begin{cases} x=1+3t \\ y=-1-t \\ z=t. \end{cases}$

l_2与平面S的交点:

$3(1+3t)-(-1-t)+t=5$、

$\Rightarrow 11t=1 \Rightarrow t=\frac{1}{11}$

$\Rightarrow B\left(\frac{14}{11}, \frac{-12}{11}, \frac{1}{11}\right)$, $\overrightarrow{AB}=\left(\frac{4}{55}, \frac{6}{55}, \frac{-6}{55}\right)$

投影AB: $\begin{cases} x=\frac{6}{5}+2t \\ y=-\frac{6}{5}+3t \\ z=\frac{1}{5}-3t \end{cases}$